NORTH BENGAL INTERNATIONAL UNIVERSITY

Lab Report : 4

TOPIC : WRITE A PROGRAM OF LECTURE-6 AND SWAP 2 NUMBERS USING 3RD VARIABLE

COURSE CODE : CSE 0612-2122

COURSE TITLE : OBJECT ORIENTED PROGRAMMING SESSIONAL.

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Write a program of Lecture-6 and swap 2 numbers using 3rd variable:

Simple example of using pointers printing the address and value:

Code:

```
#include <iostream>
using namespace std;

int main() {
    int num = 10;
    int *ptr;

    ptr = &num;      // Store the address of num in ptr

    cout << "Value of num: " << num << endl;
    cout << "Address of num: " << &num << endl;

    cout << "Value of num using pointer: " << *ptr << endl;
    cout << "Address stored in pointer: " << ptr << endl;

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

Value of num: 10
Address of num: 0x61ff08
Value of num using pointer: 10
Address stored in pointer: 0x61ff08
PS C:\Users\Lenovo\Documents\c++>
```

c++ pointer program to swap two number using 3rd variable:

Code:

```
#include <iostream>
using namespace std;

void swap(int *a, int *b) {
    int temp;
    temp = *a;
    *a = *b;
    *b = temp;
}

int main() {
    int num1, num2;
    cout << "Enter first number: ";
    cin >> num1;
    cout << "Enter second number: ";
    cin >> num2;
```

```
    // Print numbers before swapping
    cout << "Before swapping:" << endl;
    cout << "num1 = " << num1 << ", num2 = " << num2 << endl;

    // Swap the numbers
    swap(&num1, &num2);
    cout << "After swapping:" << endl;
    cout << "num1 = " << num1 << ", num2 = " << num2 << endl;

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

Enter first number: 4
Enter second number: 7
Before swapping:
num1 = 4, num2 = 7
After swapping:
num1 = 7, num2 = 4
PS C:\Users\Lenovo\Documents\c++>
```

#c++ changing value pointed by pointers:

Code:

```
#include <iostream>
using namespace std;

int main() {
    int num = 20;
    int *ptr;
    ptr = &num;

    cout << "Initial value of num: " << num << endl;

    *ptr = 50;

    cout << "Value of num after changing through pointer: " << num
<< endl;

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

Initial value of num: 20
Value of num after changing through pointer: 50
PS C:\Users\Lenovo\Documents\c++>
```

#the address of 'a' variable to teh third element of an array 'ptr':

Code:

```
#include <iostream>

int main() {
    int a = 42;
    int *ptr[5]; // Array of pointers to integers

    // Assigning the address of 'a' to the third element of 'ptr'
    ptr[2] = &a;

    // Displaying the address and value
    std::cout << "Address of 'a': " << &a << std::endl;
    std::cout << "Address stored in ptr[2]: " << ptr[2] <<
std::endl;
    std::cout << "Value pointed to by ptr[2]: " << *ptr[2] <<
std::endl;

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

Address of 'a': 0x61ff0c
Address stored in ptr[2]: 0x61ff0c
Value pointed to by ptr[2]: 42
PS C:\Users\Lenovo\Documents\c++>
```

Array to pointer to String:

Code:

```
#include <iostream>
#include <cstring>

int main() {
    // Define an array of characters
    char array[] = "Hello, world!";

    // Create a pointer to the array
    char *pointer = array;

    // Display the string using the pointer
    std::cout << "The string is: " << pointer << std::endl;

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

The string is: Hello, world!
PS C:\Users\Lenovo\Documents\c++>
```